



ARMLET

Proactive Burnout Detection for First Responders

Ahmed Faraz · Khushi Gandhi · Abby Kaur · Shrima Narasimhan

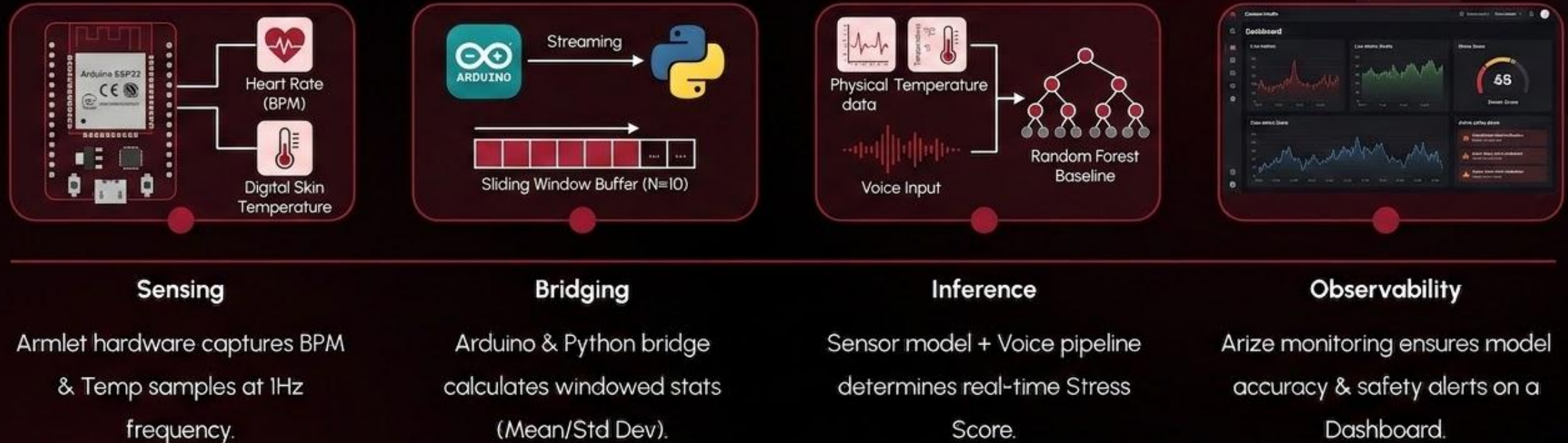
Deepgram · Arize · Ddoski's Lab

Introducing: the **ARMLET**

a wearable sensor and dashboard ecosystem designed specifically for mission-critical personnel.

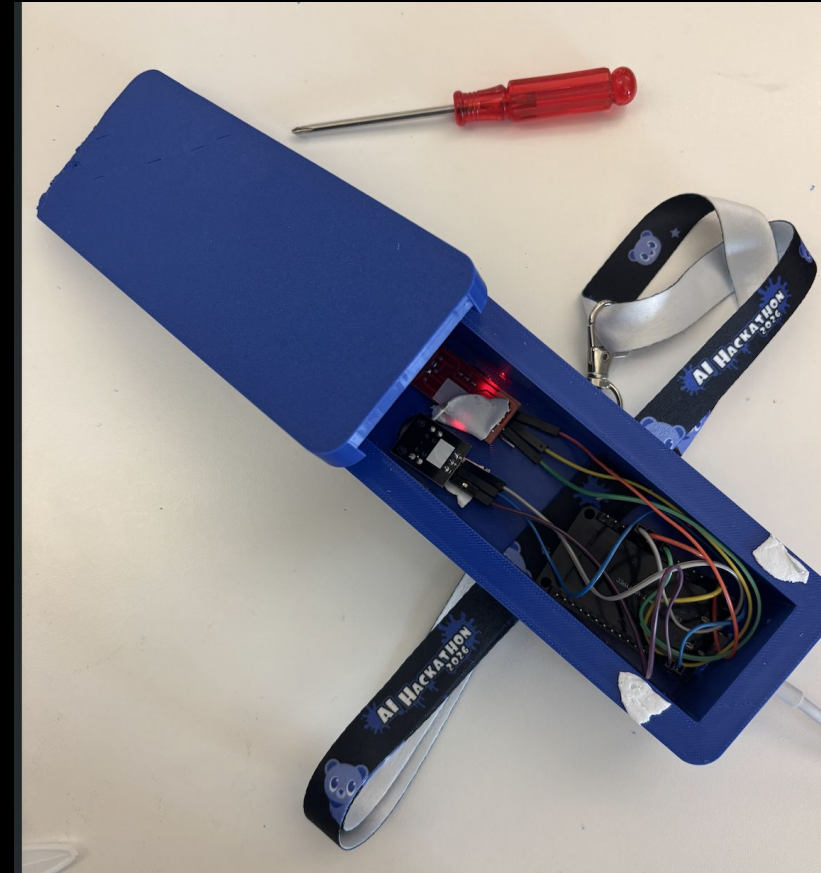


The Pipeline



The **Armlet** Hardware

- ✓ **Continuous Biometrics:** Capturing high-fidelity heart rate (BPM) and skin temperature data.
- ✓ **Edge Connectivity:** Arduino-powered streaming with low-latency serial bridging.
- ✓ **Field Durability:** Rugged design capable of sustaining 24-hour shift cycles in extreme conditions.
- ✓ **Haptic Alerts:** Direct vibration feedback for immediate physiological awareness.



AI Risk Engine



Risk Scoring

Classifying physiological state, voice tone and semantics



Thresholding

Dynamic scoring (0.0 - 1.0) with automated triggers for safety supervisors.

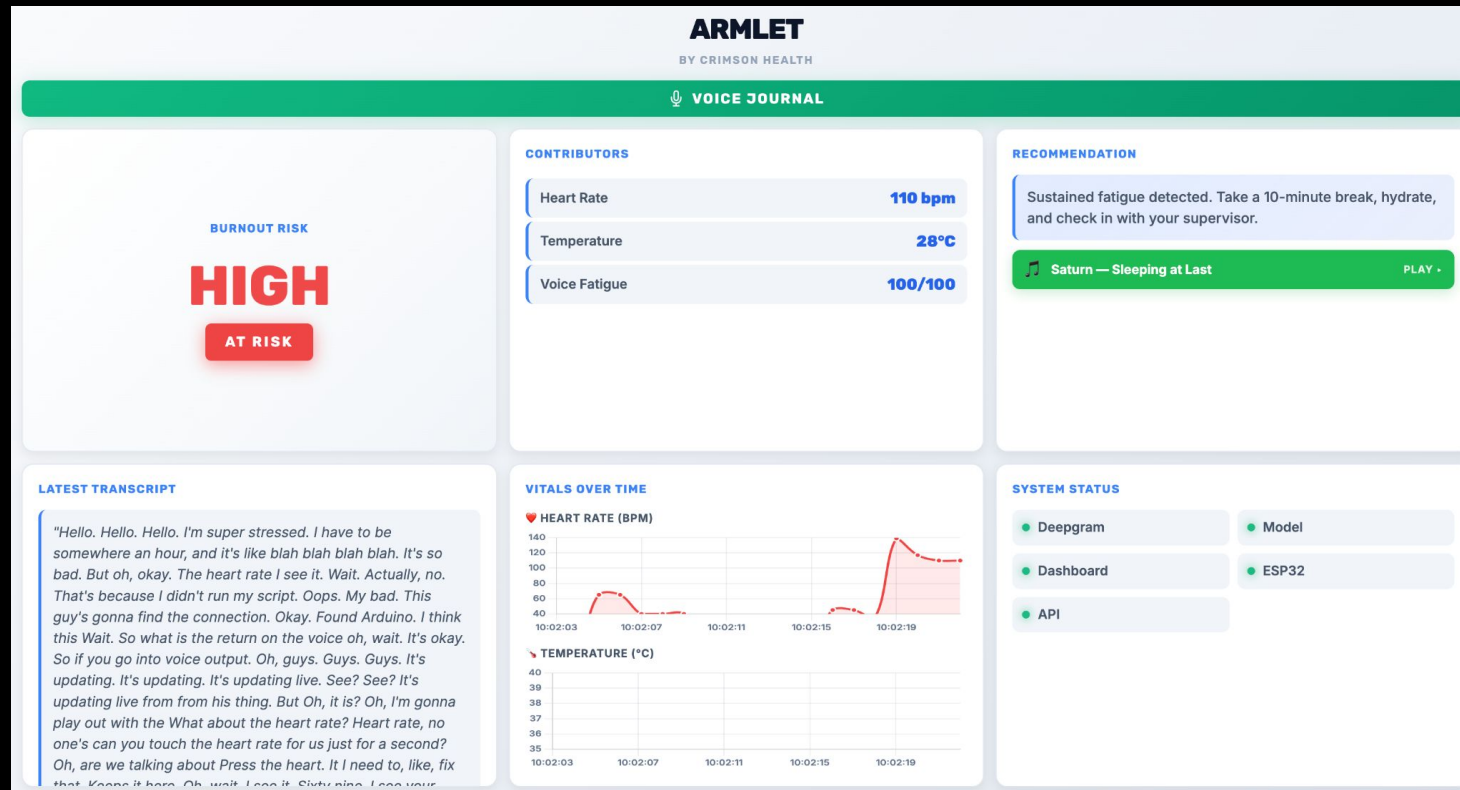


Intervention

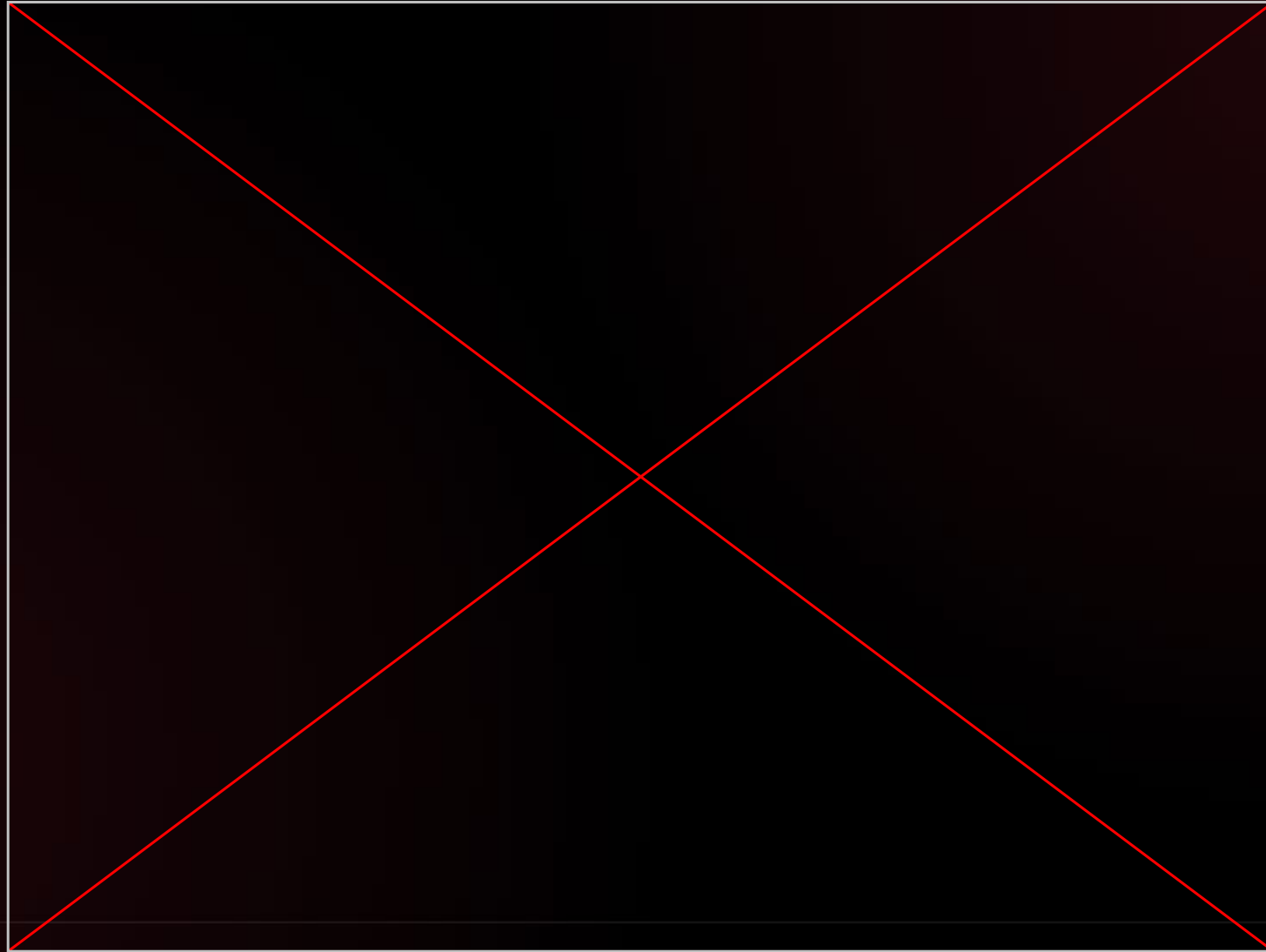
Personalized recommendations ranging from haptic breathing prompts to mandatory rest.

The Live Dashboard

- ✓ **Continuous Biometrics:** Displaying high fidelity heart rate & skin temperature data received from hardware
- ✓ **Voice Journal:** User can speak directly into dashboard UI, voicing concerns or thoughts
- ✓ **Voice Analysis:** Displays logs of recent voice input clips and analyzes their sentiment
- ✓ **Solutions:** Based on dynamic stress level, suggests stress relief techniques and links specific Spotify songs to calm the user



Demo



Strategic Impact

- ✓ **Reduce Attrition:** Protect first responders by preventing career-ending burnout.
- ✓ **Mitigate Medical Error:** Identifying fatigue before it impacts clinical judgment and patient safety.
- ✓ **Model Reliability:** Integrated Arize observability ensures the AI remains unbiased and accurate over time.
- ✓ **Fleet-Wide Wellness:** A scalable solution for fire, EMS, police, and emergency dispatch departments.



Challenges: Hardware Gaps & Async Pipelines



The Team Bottleneck

We had exactly **one hardware person** on the team; everyone else had zero hardware experience. Building Armlet required intense, continuous communication between hw and sw to ensure the data loops, socket alignments, and physical boundaries actually fit together.



A Missing Sensor, Mid-Event

Our kit arrived missing the planned motion sensor. Instead of scrambling to retrain under time constraints, we imputed mean placeholder values for motion and relied on heart rate and temp.



Dependency Dead Ends

Our first-choice pretrained tone model hit immediate version incompatibilities with our environment. We abandoned it quickly to pivot to a model that ran natively.



Hardware Availability & Simulation

Since we couldn't build out every physical node overnight, we wrote a custom dataset replayer to stream WESAD rows over local UDP sockets, faking the hardware telemetry.

What We Learned

End-to-End Sponsor Loops

- **Two-Way Deepgram Loop:** Integrated STT for voice check-in inputs and TTS for real-time, empathetic user nudges.
- **Live Arize Monitoring:** Used the SDK to track runtime inference and verify telemetry data stream health.
- **Complementary Signals:** Tested and validated that our voice tone and text models catch each other's distinct blind spots.

Cross-Discipline Execution

- **HW/SW Communication:** Synced constantly to prevent our single hardware person from working in an absolute silo.
- **Edge Partitioning:** Learned to isolate core edge-sensing streams from higher-level dashboard platforms.
- **Agile Pivots:** Learned to drop breaking dependencies quickly instead of wasting critical build hours.

What's Next

Trust & Data Security

- As Armlet handles sensitive biometric and voice data, productionizing this would require end-to-end encryption
- processing and discarding raw audio rather than storing it), and clear consent and control for the wearer

- **Production Grade Acoustic Modeling** - trained on industry-level first responder data rather than generic:
- **Environmental Sensing:** Adding a humidity sensor alongside temperature and motion would let the system distinguish environmental heat stress

Thank you!

Questions?

Protecting those who protect us.

